MIL-T-16652(BuMed)
1 December 1951
SUPERSEDING
MDTS 1729
14 October 1944

MILITARY SPECIFICATION

TOURNIQUET, PNEUMATIC

1. SCOPE

1.1 This specification covers pneumatic tourniquets for use in controlling blood supply to the extremities during surgery.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

NN-B-601	Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621	Boxes, Wood, Nailed and Lock-Corner.
NN-B-631	Boxes; Wood, Wirebound (for Domestic Shipment).
ZZ-R-601	Rubber-Goods; General Specifications. (Method of Physical Tests and Chemical Analyses).
LLL-B-631	Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636	Boxes; Fiber, Solid (for Domestic Shipment).

Military

JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-B-107	Boxes, Wood, Wirebound (Overseas Type).
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-140	Packaging and Packing for Overseas Shipment - Adhesives, Water-Resistant, Case-Liner.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking of Shipments.

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DRAWINGS

Armed Services Medical Procurement Agency

1056 Insignia.
1801 Tourniquet, Pneumatic.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Envelope and belt. - Envelope and belt shall be cotton balloon cloth made with a 5 by 5 basket-weave, and having a minimum weight of 5.2 ounces per square yard. Cloth shall have a thread count of 100 threads per inch in the warp and in the filling (minus tolerance of 2 threads per inch in each direction) and a breaking strength of 140 pounds in the warp and in the filling (minus tolerance of 5 pounds in each direction). Cloth shall be coated with black or green, natural, or synthetic rubber of such properties as to meet the requirements hereinafter specified.

3.1.1.1 Reinforcing material. - Envelope reinforcing material shall be stainless steel wire having an outside diameter of approximately 0.045 inch and a Rockwell hardness of C-35 C-40.

3.1.2 Outlet tube. - Outlet tube shall be nickel plated brass.

3.1.3 Buckle. - Buckle shall be bronze, brass, steel or aluminum, treated to prevent corrosion.

3.1.4 Tubing. - Tubing shall be black, or green, natural, or synthetic rubber of such properties as to meet the requirements hereinafter specified.

3.1.5 Inflation pump. - Inflation pump shall be brass, chrome or nickel plated, suitable for purpose intended.

3.1.6 Case. - Case shall conform to requirements of latest purchase description for Stock No. 9-129-950, Case, Medical Field Kit, Rigid, No. 3, Empty, except that material shall be either aluminum alloy or steel, and requirements for partition shall be omitted. If steel is used, body and cover of case shall be seamlessly die-drawn of 0.0359 inch sheet steel and trimmed.

3.1.6.1 Pressure gage receptacle. - Fitting within the case for use as receptacle for the pressure gage shall be of material suitable for securely holding gage and to provide cushioning for the gage.

3.2 Construction. - Pneumatic tourniquet shall consist essentially of an inflation envelope with belt and buckle, and an inflation-deflation assembly consisting of a hand pump, pressure gage, and tubing. Dimensions and design of tourniquet components shall be as shown on Drawing 1801.

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3.2.1 Envelope. - The envelope shall be tubular, molded into a circular shape. In cross-section, the shape of the envelope shall approximate a semi-circle with the flat side outward. The flat side shall be reinforced with steel wire, as shown on drawing, so that when envelope is inflated this outer side shall maintain an approximately flat surface. The inner side (semi-circular arc) shall be so constructed that when the tourniquet is applied to an arm or leg no wrinkles shall develop where the envelope is in contact with the arm or leg.

3.2.2 Outlet tube. - The envelope shall be inflated through an integrally molded outlet tube, placed 3 inches from the end of the envelope (and to which the buckle is attached) on the center line. Outlet tube shall be constructed to fit tubing as specified below.

3.2.3 Belt. - The belt shall be securely attached by means of a cement, containing a vulcanizing agent, to the outlet tube end of the envelope for a distance of approximately 7-1/2 inches (measured from the attached buckle).

3.2.4 Inflation deflation system. - One end of a 4 foot length of rubber tubing (of 3/16 inch inside diameter and 3/32 inch wall) shall be attached to the outlet end, the other end fitted to a side-arm nipple of the pressure gage. Each end of a length of tubing approximately 18 inches long (of 3/16 inch inside diameter and 3/32 inch wall) shall be provided with a knurled coupling as shown on drawing. One coupling shall fit the bottom threaded portion of the pressure gage. The other coupling (containing a Schrader valve core) shall fit the distal end of a hand pump. Two leather washers shall be placed on this coupling (between coupling and pump).

3.2.4.1 Pressure gage. - The pressure gage shall be a Bourdon tube gage as shown on Drawing 1801. Face of gage shall be marked with black numbers to indicate pressures from zero to 600 mm. of mercury. To prevent exceeding the arm and leg pressure limits there shall be a red line marker on the face of the gage at the 260 mm. of mercury level and at the 520 mm. of mercury level with the wording "Do Not Exceed" under each red line and the word "Arm" under the 260 mm. red line and the word "Leg" under the 520 mm. red line. The gage shall accurately indicate pressures with deviations not exceeding 1 percent in the range between 250 and 525 mm. of mercury and 2 percent over the remainder of the scale.

3.2.5 Leakage. - The tourniquet shall be so constructed that when in operation all fittings and closures shall be airtight. Tourniquet shall be capable of passing leakage test specified in 4.3.1.

3.2.6 Case. - Case construction and finish shall be in accordance with latest purchase description for Stock No. 9-129-950, Case, Medical Field Kit Rigid, No. 3, Empty, except that requirements for partition shall be omitted, and inside depth of the case shall measure 2-7/8 inches.

3.2.6.1 Pressure gage receptacle. - Receptacle within the case shall be so constructed that it will securely retain the pressure gage and adequately protect it from damage.

3.3 Sterilization. - Rubber parts of tourniquet shall be capable of withstanding sterilization by boiling water or by standard cold sterilizing solutions.

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3.4 Aging properties of rubber parts. - After subjection to accelerated aging test specified in 4.3.2, rubber parts shall show no evidence of becoming tacky, brittle, or cracking.

3.5 Instruction sheet. - Securely mounted on the inside surface of the case cover shall be a kraft-colored instruction sheet, not less than 5 by 7 inches in size, printed in easily legible black type with complete instructions for use of the pneumatic tourniquet and bearing adequate illustrations of use.

3.6 Identification marking.

3.6.1 Tourniquet. - The belt of the tourniquet shall be marked with the name or registered trade-mark of the manufacturer.

3.6.2. Case. - The outside surface of the case cover shall be centrally stenciled with a caduceus between the letters "U" and "S," conforming to Drawing 1056. The caduceus shall be approximately 2-3/4 inches in height and the letters "U" and "S" approximately 3/4 inch in height. The cover shall also be stenciled with the nomenclature and stock number of the tourniquet. All marking on case shall be accomplished using a black lusterless enamel.

3.7 Workmanship. - Workmanship shall be first class throughout. Tourniquets shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. - Unless otherwise specified (see 6.1), samples shall be submitted in accordance with instructions furnished by the procuring activity. In addition, inspection at contractor's facility by inspector shall be conducted in accordance with Standard MIL-STD-105.

4.2 Inspection. - Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.

4.3.1 Leakage. - The tourniquet shall be completely assembled with gage, pump and tubing, and shall be inflated to a pressure of 600 mm. of mercury. Master valve on the gage shall then be closed and pump and tubing connection (coupled to bottom of gage) shall be removed. Tourniquet (with tubing and gage attached) shall remain inflated for a period of 2 hours, after which time it shall not lose more than 50 mm. of mercury.

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4.3.2 Accelerated aging. - The rubber components (with metal parts where attached) of the tourniquet shall be subjected to the air oven test specified in Specification ZZ-R-601 for a period of 7 days at 70° ± 1° C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - One tourniquet, as specified, constitutes one unit. Each unit shall be packaged in a solid fiberboard or double-faced corrugated fiberboard carton, of appropriate size and design, constructed in accordance with Specification LLL-B-631 or LLL-B-636. Each carton shall be adequately secured to prevent accidental opening.

5.2 Packing. -

5.2.1 For domestic shipment. - Twelve unit cartons (12 units) shall be overpacked in an exterior container constructed in accordance with Specification NN-B-601, NN-B-621, or NN-B-631.

5.2.2 For overseas shipment. - Twelve unit cartons (12 units) shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with Specification JAN-P-105, JAN-P-106, or JAN-B-107. Closure and strapping shall comply with appendix of applicable specification. Each box shall be lined with a waterproof barrier (bag) conforming to Specification JAN-P-125. Closures shall be completely sealed by a continuous seam of water-resistant adhesive at least 3/4 inch wide and conforming to Specification JAN-P-140.

5.2.3 Packing variation permitted. - If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in an exterior container of suitable size and design and so constructed as to insure safe delivery to specified destination.

5.3 Marking.

5.3.1 Interior packages. - Each unit carton shall be marked with stock number, nomenclature, unit, quantity, and name of contractor.

5.3.2 Shipping container. - Shipping container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sampling required (see 4.1).
(c) Inspection required (see 4.2).
(d) Place of final inspection (see 4.2).
(e) Whether domestic or overseas shipment is required (see 5.2).

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6.2 This specification covers the following item appearing in the Armed Services Catalog of Medical Materiel:

3-830-400 Tourniquet, Pneumatic.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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